SPE-229116-MS

Ensuring Well Integrity in Carbon Storage: Modeling CO_2 Flow, Temperature and Pressure Behavior, and Stress Risks

Yongfeng Kang, Zhengchun Liu, and Mihai Fleica, Halliburton, Houston, TX, USA

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

Abstract

Carbon Capture, Utilization, and Storage (CCUS) is pivotal in achieving global net-zero or negative carbon emissions. This study focuses on the modeling of downhole pressure and temperature (P/T) profiles during CO_2 injection, which is essential for the design of new wells or the evaluation of existing wells to ensure long-term integrity in carbon storage applications.

This work employs Wagner's Equation of State (EOS), consistent with NIST standards, to compute precise thermodynamic properties and phase behavior of CO_2 across varying P/T conditions. A transient, multiphase flow model is developed to simulate CO_2 injection dynamics, capturing phase transitions (vapor/liquid/supercritical) and their impact on downhole conditions. The model is integrated into an advanced well design software, enabling simulation of transient thermal and hydraulic behavior, as well as complex stress analysis for tubular integrity. This integrated approach provides a comprehensive view of downhole thermal and mechanical behavior during CO_2 injection, supporting well design and operational planning.

Simulation results provide key operational parameters such as CO_2 flowrate, velocity, density, quality, multiphase flow regions, and liquid holdup, along with detailed downhole P/T profiles and their historical evolution. These P/T profiles are used as inputs for wellbore stress analysis to assess both newly designed and existing wells for their suitability in CO_2 injection operations. Stress analysis considers safety factors and highlights thermal effects on tubular components, including risks of tensile failure due to cooling and potential low-temperature-induced material degradation—especially critical for older wells. The study found that CO_2 injection significantly reduces downhole temperatures compared to conventional fluids, potentially leading to tensile failure in tubing and casing. Additionally, detection of solid phase CO_2 formation triggers operational warnings due to its potential to cause tubing erosion. These insights inform injection parameter adjustments and highlight the need for careful thermal management during CO_2 injection. The modeling approach serves as a valuable tool in predicting and mitigating mechanical integrity risks during CCUS operations.

This paper presents a novel integration of accurate CO_2 thermodynamic modeling and transient flow simulation with tubular stress analysis to support CCUS well design and operation. It enhances the understanding of downhole P/T behavior during CO_2 injection and provides a predictive framework for evaluating well integrity, offering practical value in repurposing existing wells for carbon storage.

Introduction

Carbon Capture, Utilization, and Storage (CCUS) has become one of the leading solutions in the fight against climate change. As the world moves toward net-zero carbon emissions, technologies that capture and store carbon dioxide (CO_2) deep underground are becoming essential. One of the most important parts of CCUS is the safe and reliable injection of CO_2 into underground rock formations. For this process to work properly and safely, scientists and engineers must understand what happens to the CO_2 after it is injected—especially how its pressure and temperature (P/T) behave inside the wellbore.

Modeling these downhole P/T conditions is a complex task, but it is necessary to make sure that the equipment used, like steel tubing and casings, can handle the forces they will experience during the CCUS injection and storage lifecycle. These models help decide how to build the wells and how to operate them, so they don't fail over time. Without accurate models, operators risk damaging wells or releasing CO_2 into the atmosphere—defeating the purpose of storage entirely.

CO_2 behaves very differently from other fluids commonly used in oil and gas operations. Depending on the temperature and pressure, CO_2 can exist as a gas, a liquid, or a supercritical fluid (a state that is neither gas nor liquid but has properties of both), per the CO_2 phase diagram in Fig. 1. These changes in state are important because they affect how CO_2 flows, how much space it takes up, how much it cools or heats the surrounding materials, and what kind of stress it places on the steel pipes in the well and induced due to the temperature change.

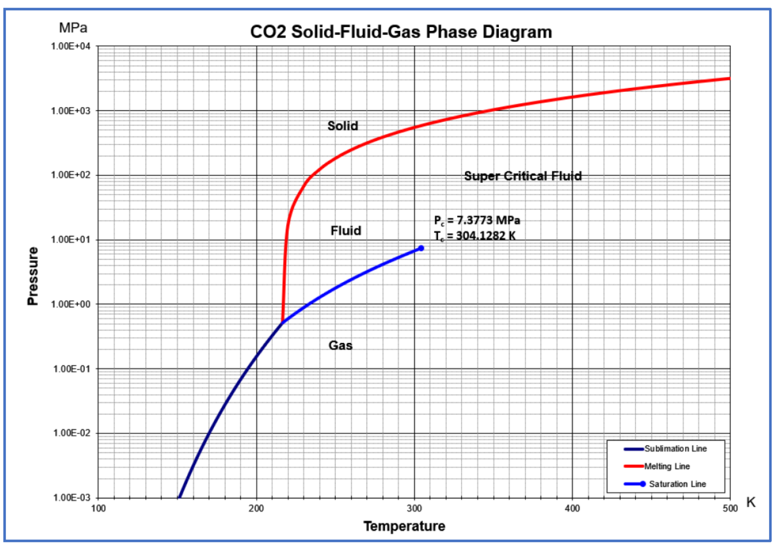

Figure 1—Phase diagram of pure CO_2

For example, when CO_2 changes from liquid to gas or enters a supercritical state, its volume and temperature can change rapidly. This often results in rapid cooling of the wellbore, which can shrink the steel pipes and cause them to crack or lose strength over time. One study showed that CO_2 injection can cause large drops in temperature, which may increase stress in the tubing and lead to damage, especially in older wells (Yang et al., 2024).

To model how CO_2 behaves underground, scientists use different equations to model it, in which the equations of state (EOS) is a special type of equation. These equations calculate the physical properties of a substance under different P/T conditions. One of the most accurate EOS for modeling CO_2 is the Span-Wagner Equation of State. This equation was specifically designed to handle the complex behavior of pure CO_2 across all its phases—gas, liquid, and supercritical (McKay et al., 2022) in the range of Triple-Point to 1100K for temperature and pressure up to 800 MPa (Span and Wagner, 1996). The Span-Wagner EOS is widely accepted by engineers and researchers because it offers very high precision, especially when compared to older cubic EOS models. It has been used in studies across the world, including for pipeline transport, wellbore simulations (Bai, 2024).

When this equation is built into computer models, it helps simulate what happens inside the well during CO_2 injection. For example, researchers have used it to predict how CO_2 temperature changes over time, what kind of pressure it exerts at different depths, and how quickly it might cool surrounding metal parts (Mao et al., 2023).

But thermodynamics properties alone are not enough. Engineers also need to understand how CO_2 flows in the well, so transient multiphase flow models are needed. These are simulations that show how CO_2 moves through the well over time, including when it exists in more than one phase at once—like part gas and part liquid. Multiphase flow modeling is especially important for CCUS because CO_2 often flows as a mix of gas and liquid in different parts of the well. In subsea or offshore injection systems, this is even more critical due to rapid environmental temperature drops (Lien, 2023). These models help engineers plan for different flow conditions, determine where liquid may collect, and predict if and where CO_2 might turn into dry ice (solid CO_2), which can clog or erode the tubing as well as casing/liner. A study used these models to simulate pressure and temperature along a CO_2 injection well, considering how the gas and liquid interacted as they moved. These simulations revealed zones of rapid cooling and pressure changes, offering insights into the kind of stress the well would experience under real operating conditions (Andrianov et al., 2023).

When CO_2 cools the tubulars of the well, the metal contracts and results in tensile stress concentration. If the change of the temperature is too large or happens too quickly, the metal can stretch or crack due to the induced stress concentration. Over time, this can weaken the well integrity. To model a CO_2 injection well integrity, the stress analysis is required. This involves taking the pressure and temperature data from the thermodynamic and flow simulations and using them to figure out how much stress the steel tubing and casing will experience to evaluate whether these tubulars could handle such stresses. Engineers use safety factors and models to make sure the materials can handle the expected conditions (Liu and Samuel, 2023). Some recent models also consider how repeated cycles of heating and cooling can lead to long-term fatigue damage (Lv et al., 2025). In offshore projects, such as those in the North Sea or the Aurora Storage Site, engineers apply this combined modeling approach to balance injection rates while aiming to enhance storage efficiency while minimizing leakage risk (Alali, 2023).

As the scenarios get more and more complex, an integrated comprehensive solution is needed that could do not only the calculation of the transient thermodynamic calculation of the CO_2 injection in the well but also perform the stress analysis that the wellbore tubulars will experience and well integrity analysis to ensure the wellbore will still be safe during its service lifecycle. That is the purpose of this study in order to not only guide how to build new wells but also help evaluate whether old oil and gas wells can be reused for CO_2 storage, or guide how to best adjust the operation parameters to best reuse old oil and gas wells for CCUS purpose.

In summary, the modeling of pressure and temperature during CO_2 injection is a critical part of designing safe and reliable CCUS wells. By combining accurate thermodynamic models like the Span-Wagner EOS, advanced multiphase flow simulations, and mechanical stress analysis, it could predict and prevent failure in injection wells. This helps make carbon storage more efficient, safer, and more cost-effective – helping to move the world one step closer to climate goals.

Modeling of CO_2 properties

This section will talk about the modeling method used in this study. The properties calculated are all compared with NIST (National Institute of Standards and Technology) Chemistry web-book properties and have a very good match.

CO_2 Properties Modeling with Span-Wagner EOS

Most of the CO_2 properties are calculated with the Span-Wagner EOS model (Span and Wagner, 1996), which is a widely accepted method to calculate the properties of pure CO_2 accurately, or it should say that the Span-Wagner EOS is the standard calculation method now where the NIST also applies this EOS model in their modeling as standard reference (NIST, 2025).

The set of Span-Wagner EOS (Span and Wagner, 1996) is expressed in the form of a fundamental equation explicit in the Helmholtz free energy and it is capable to calculate the properties of CO_2 covering the fluid region from the triple point temperature to 1100 K at pressures up to 800 MPa. It was stated that in the most important region up to pressures of 30 MPa and up to temperatures of 523 K technically, the estimated uncertainty of the equation ranges from ± 0.03% to ± 0.05% in the density, ± 0.03% to ± 1% in the speed of sound, and ± 0.15% to ± 1.5% in the isobaric heat capacity. This EOS is also claimed by Span and Wagner that at least for the basic properties such as pressure, fugacity, and enthalpy, the EOS can be extrapolated up to the limits of the chemical stability of carbon dioxide. This way, the EOS could calculate very accurate properties for the pure CO_2 for thermodynamic pressure, temperature and density, specific isobaric heat capacity, specific isochoric heat capacity, speed of sound, entropy, enthalpy, internal energy, Joule-Thomson coefficient, saturated liquid heat capacity, fugacity, etc. Independent equations for the vapor pressure and for the pressure on the sublimation and melting curve, for the saturated liquid and vapor densities, and for the isobaric ideal gas heat capacity are also included in their model.

A brief summary of the Span-Wagner EOS for CO_2 and the independent equations for the associated properties are given as below, where for much more details, especially the detailed coefficients, it would recommend referring to the original paper of Span-Wagner (1996):

The EOS is expressed in the form of Helmholtz energy *A(ρ, T)* with two independent parameters, density ρ and temperature T. The dimensionless Helmholtz energy $\varnothing = A/(RT)$ is expressed with a part of ideal-gas behavior $\varnothing^o$ and a part of the residual fluid behavior $\varnothing^r$ as below:

$$\varnothing(\delta, \tau) = \frac{A(\rho, T)}{RT} = \varnothing^o(\delta, \tau) + \varnothing^r(\delta, \tau) \tag{1}$$

Where the part of ideal gas behavior, $\varnothing^\circ(\delta, \tau)$, is given as:

$$\varnothing^o(\delta, \tau) = ln(\delta) + a_1^o + a_2^o\tau + a_3^o ln(\tau) + \sum_{i=4}^{8} a_i^o ln\left[1 - exp\left(-\tau\theta_i^o\right)\right] \tag{2}$$

And the part of the residual fluid behavior, $\varnothing^r(\delta, \tau)$, is given as:

$$\varnothing^r(\delta, \tau) = \sum_{i=1}^{7} n_i \delta^{d_i} \tau^{t_i} + \sum_{i=8}^{34} n_i \delta^{d_i} \tau^{t_i} e^{-\delta^{c_i}} + \sum_{i=35}^{39} n_i \delta^{d_i} \tau^{t_i} e^{-a_i(\delta-\varepsilon_i)^2 - \beta_i(\tau-\gamma_i)^2} + \sum_{i=40}^{42} n_i \Delta^{b_i} \delta e^{-C_i(\delta-1)^2 - D_i(\tau-1)^2} \tag{3}$$

$$\Delta = \left\{(1-\tau) + A_i\left[(\delta-1)^2\right]^{\frac{1}{2\beta_i}}\right\}^2 + B_i\left[(\delta-1)^2\right]^{a_i} \tag{4}$$

Where $\delta = \frac{\rho c}{\rho}$ is the reduced density with the critical density, ρ_c

$\tau = \frac{T}{T_c}$ is the inverse reduced density with the critical temperature, T_c

R is the specific gas constant, $R = \frac{R_m}{M} = (0.1889241 \pm 0.0000116) kJ \Big/ (kg \cdot K)$

R_m is the molar gas constant, $R_m = (8.314510 \pm 0.000210)\ J/(mol \cdot K)$

M is the CO_2 molar mass, $M = (44.0098 \pm 0.0016)\ g/mol$

a_i^o, θ_i^o are the coefficients for ideal gas behavior part, values referring to original paper of Span-Wagner (1996) n_i, d_i, t_i, c_i, α_i, β_i, γ_i, ϵ_i, a_i, b_i, A_i, B_i, C_i, D_i are the coefficients for residual fluid behavior part, values referring to original paper of Span-Wagner (1996)

All the thermodynamic properties of pure CO_2 can be obtained by combining partial derivatives of the equations (1), (2), (3). For example, the isobaric heat capacity could be calculated with:

$$\frac{c_p(\delta,\tau)}{R} = -\tau^2\left(\phi^o_{\tau\tau} + \phi^r_{\tau\tau}\right) + \frac{\left(1+\delta\,\phi^r_\delta - \delta\tau\,\phi^r_{\delta\tau}\right)^2}{1+2\delta\,\phi^r_\delta + \delta^2\,\phi^r_{\delta\delta}} \tag{5}$$

Another example is that the pressure is calculated from below equation:

$$\frac{P(\delta,\tau)}{\rho RT} = 1+\delta\,\phi^r_\delta \tag{6}$$

The other independent equations are also given by Span and Wagner (1996).

For example, the melting pressure is calculated as:

$$\frac{P_m}{P_t} = 1 + a_1\left(\frac{T}{T_t} - 1\right) + a_2\left(\frac{T}{T_t} - 1\right)^2 \tag{7}$$

Where P_m is the melting pressure

P_t is the triple point pressure, = 0.51795 MPa

T_t is the triple point temperature, = 216.592 K

$a_{1,2}$ are coefficients

The sublimation pressure is calculated as:

$$ln\left(\frac{p_{sub}}{P_t}\right) = \frac{T}{T_t}\left\{a_1\left(1-\frac{T}{T_t}\right) + a_2\left(1-\frac{T}{T_t}\right)^{1.9} + a_3\left(1-\frac{T}{T_t}\right)^{2.9}\right\} \tag{8}$$

Where p_{sub} is the sublimation pressure

P_t is the triple point pressure, = 0.51795 MPa

T_t is the triple point temperature, = 216.592 K

$a_{1,2,3}$ are coefficients

The saturated liquid density is calculated as:

$$ln\left(\frac{\rho'}{\rho_c}\right) = \sum_{i=1}^{4} a_i\left(1-\frac{T}{T_c}\right)^{t_i} \tag{9}$$

Where ρ' is the saturated liquid density

ρ_c is the critical point density, = 467.6 kg/m^3

T_c is the critical point temperature, = 304.1282 K

a_i are coefficients

t_i are coefficients

The saturated vapor density is calculated as:

$$ln\left(\frac{\rho''}{\rho_c}\right) = \sum_{i=1}^{5} a_i\left(1-\frac{T}{T_c}\right)^{t_i} \tag{10}$$

Where ρ'' is the saturated vapor density

ρ_c is the critical point density, =467.6 kg/m^3

T_c is the critical point temperature, = 304.1282 K

a_i are coefficients

t_i are coefficients

The density of the saturated liquid and the saturated vapor at the triple point are calculated from the above equations:

$$\rho'_{tl} = (1178.53 \pm 0.18) kg/m^3 \tag{11}$$

$$\rho''_{tg} = (13.7614 \pm 0.0034) kg/m^3 \tag{12}$$

The critical point temperature and pressure are given as:

$$T_c = (304.1282 \pm 0.015)K \tag{13}$$

$$P_c = (7.3773 \pm 0.0030)MPa \tag{14}$$

In summary, the CO_2 properties of density, specific internal energy, enthalpy, Cv, Cp, speed of sound, Joule-Thomson coefficient, saturation density (liquid, and vapor), melting pressure and temperature, sublimation pressure and temperature are calculated with Span-Wagner EOS (Span and Wagner, 1996). Only a few of the equations provided in the original paper are briefly included in this paper due to the complexity of the equations, coefficients, and derivatives, so readers are required to read the original paper by Span and Wagner if they want to know more details.

Thermal Conductivity of CO_2

The thermal conductivity property of CO_2 is calculated with the model proposed by Scalabrin et al. (2006) which is in a multiparameter format through the application of an optimization technique of the functional form and valid for temperature from triple point to 1000 K and pressure up to 200 MPa with an average absolute deviation of 1.19% on the selected 1407 primary data points. The equation is given as follows (Scalabrin et al., 2006), where the density for the reduced density as an input is calculated from the above Span-Wagner EOS:

$$\lambda_r(T_r, \rho_r) = \sum_{i=1}^{3} n_i T_r^{g_i} \rho_r^{h_i} + e^{-5\rho_r^2} \sum_{i=4}^{10} n_i T_r^{g_i} \rho_r^{h_i} + n_c \lambda_{r,ce}(T_r, \rho_r) \tag{15}$$

With the critical term:

$$\lambda_{r,ce}(T_r, \rho_r, \overline{a}) = \frac{\rho_r \exp\left\{\frac{\rho_r^{a_i}}{a_i} - \left[a_2(T_r - 1)\right]^2 - \left[a_3(\rho_r - 1)\right]^2\right\}}{\left(\left[\left\{\left(1\frac{1}{T_r}\right) + a_4\left[(\rho_r - 1)^2\right]^{1/2a_5}\right\}^2\right]^{a_6} + \left\{\left[a_7(\rho_r - \alpha)\right]^2\right\}^{a_8}\right)^{a_9}} \tag{16}$$

$$\alpha = \alpha(T_r) = 1 - a_{10} arcosh\left\{1 + a_{11}\left[(1 - T_r)^2\right]^{a_{12}}\right\} \tag{17}$$

Where the coefficients are given in the tables below:

Viscosity of CO_2

The viscosity property of CO_2 is calculated with the model proposed by Laesecke and Muzny (2017) which has the coverage for temperature from 100 K to 2000 K for gaseous CO_2, and from 220 K to 700 K with pressure along the melting line up to 8000 MPa for compressed and supercritical liquid states. This model is also implemented in their NIST Reference Fluid Thermodynamic and Transport Properties Database (REFPROP). The authors did comparisons with experimental data, and concluded that the result has an uncertainty of 0.2 % in the temperature range from 200 K to 700 K, 0.6 % to 100 K and 1500 K, and 1 % to 2000 K.

The viscosity correlation for CO_2 was formulated in the following general structure (Laesecke and Muzny, 2017):

$$\eta(T, \rho) = \eta_0(T) + \rho \cdot \eta_1(T) + \Delta\eta_r(T, \rho) + \Delta\eta_c(T, \rho) \tag{18}$$

Where η is the dynamic viscosity

T is the absolute temperature

ρ is the density

η_0 (T) is the viscosity in the limit of zero density

η_1 (T) is the linear-in-density viscosity coefficient

$\Delta\eta_r$ (T,ρ) is the temperature- and density-dependent residual viscosity

$\Delta\eta_c\ (T,\rho)$ is the term for the enhancement of the viscosity near the gas-liquid critical point.
The four contributions to the above formulation are detailed as below (Laesecke and Muzny, 2017):

$$\eta_0(T) = \frac{1.0055 \cdot \sqrt{T}}{\left(a_0 + a_1 T^{\frac{1}{6}} + a_2 \cdot exp\left(a_3 \cdot T^{\frac{1}{3}}\right) + \frac{a_4 + a_5 T^{\frac{1}{3}}}{exp\left(T^{\frac{1}{3}}\right)} + a_6 \sqrt{T}\right)} \tag{19}$$

Where the parameters a_i are given in table 3 below:

Table 1—Coefficient for thermal conductivity equation

i	g_i	h_i	n_i	*i*	g_i	h_i	n_i	*i*	g_i	h_i	n_i
1	0.0	1	7.69857587	5	1.0	2	16.3890156	9	3.5	0.0	-0.171779133
2	0.0	5	0.159885811	6	1.5	0	3.69415242	10	5.5	0.0	0.004330433
3	1.5	1	1.56918621	7	1.5	5.0	22.3205514				
4	0.0	1	-6.73400790	8	1.5	9.0	66.142095			n_c	0.775547504

Table 2—Coefficient for critical term for thermal conductivity equation

i	a_i	*i*	a_i	*i*	a_i	*i*	a_i	*i*	a_i	*i*	a_i
1	3.0	3	0.94604	5	0.30	7	0.33791	9	0.79857	11	0.02
2	6.70697	4	0.30	6	0.39751	8	0.77963	10	0.90	12	0.2

Table 3—Coefficient for critical term for thermal conductivity equation

i	a_i	*i*	a_i
0	1749.35489318835000	4	-269503.24793356900000
1	-369.06930000712800	5	73145.02153182600000
2	5423856.34887691000000	6	5.34368649509278
3	-2.21283852168356		

$$\eta_1(T) = \eta_0(T) \cdot B_\eta^*(T^*) \cdot \sigma^3 \cdot \frac{N_A}{M} \tag{20}$$

$$B_\eta^*(T^*) = b_0 + \sum_{i=1}^{8} \frac{b_i}{(T^*)^{t_i}} \tag{21}$$

Where $B_\eta^*(T^*)$ is the temperature-dependent reduced second viscosity virial coefficient,
σ is the length scaling parameter, $\sigma = 0.378421\ nm$
N_A is the Avogadro constant, $NA = 6.022140857(74) \times 10^{23}\ mol^{-1}$
M is the CO_2 molar mass, $M = 44.0095 \times 10^{-3}\ kg/mol$
T^* is the reduced temperature and $T^* = \frac{T}{\varepsilon / k_B}$,
ε/k_B is the energy scaling parameter, and $\frac{\varepsilon}{k_B} = 200.760\ K$
b_i and t_i are parameters and listed at table 4 below.

Table 4—Coefficient for critical term for thermal conductivity equation

i	b_i	t_i	*i*	b_i	t_i
0	-19.57288100	—	5	2491.65970000	1.25
1	219.73999000	0.25	6	-787.26086000	1.5
2	-1015.32260000	0.5	7	14.08545500	2.5
3	2471.01250000	0.75	8	-0.34664158	5.5
4	-3375.17170000	1			

$$\Delta\eta_r(T,\rho)=\eta_{tL}\cdot\left(c_1\cdot T_r\cdot\rho_r^3+\frac{\left(\rho_r^2+\rho_r^\gamma\right)}{\left(T_r-c_2\right)}\right) \tag{22}$$

Where η_{tL} is a dimensioning factor,
c_1, c_2, γ are coefficients, and given in table 5 below,
T_r is the reduced temperature, and $T_r = T/T_t$
T_t is the CO_2 triple-point temperature
ρ_r is the reduced density, and $\rho_r = \rho/\rho_{tL}$
ρ_{tL} is the density of the liquid phase CO_2 at triple point

Table 5—coefficients for equation (22)

c_1	0.360603235428487	c_2	0.121550806591497	γ	8.062827374812770

The dimensioning factor, η_{tL} is calculated from:

$$\eta_{tL}=\frac{\rho_{tL}^{2/3}\sqrt{R\cdot T_t}}{M^{1/6}\cdot N_A^{1/3}} \tag{23}$$

Where R is the molar gas constant, and R = 8.314459848 *J/mol*,

$$\Delta\eta_c(T,\rho)=\left[\eta_0(T)+\rho\cdot\eta_1(T)+\Delta\eta_r(T,\rho)\right]\cdot exp\left[z_\eta\cdot H\right] \tag{24}$$

Where z_η is the universal critical exponent,
H is the crossover function

It needs to be mentioned that CO_2 viscosity is experimentally observed to be singular at the critical point, so the critical point viscosity enhancement is applied with the above part of $\Delta\eta_c$ (T,ρ). Outside of the small region 304.13 K ≤ T ≤ 306 K with pressures 7.375 MPa ≤ P ≤ 7.677 MPa, this contribution is negligible.

Surface Tension of CO_2

The viscosity CO_2 property is calculated with the model proposed by Mulero et al. (2012). The surface tension calculation is valid along the saturation line (liquid-vapor boundary line) from the triple point to the critical point as physically the surface tension effect only exists along this saturation line. The reported mean absolute percentage deviation (MAPD) of 5.10% from the 133 measurement data points, by comparing with the 6.34% MAPD of REFPROP V9.0 of NIST.

The CO_2 surface tension equation is formulated in the form below (Mulero et al. 2012):

$$\sigma(T)=\sigma_0\left(1-\frac{T}{T_c}\right)^{n_0} \tag{25}$$

With substance specific parameters for CO_2 and equation coefficients are listed below 6 table.

Table 6—coefficients for equation (25)

σ_0 (mN/m)	0.07863	n_0 (MPa)	1.254

Temperature and Pressure, and Stress Analysis with CO_2 Injection

Once the properties of CO_2 modeling are completed, the step comes to apply the properties to do the simulation of the CO_2 injection in the well. A comprehensive framework of coupled transient thermal and fluid flow in well has already been developed in our advanced tubing and casing design tool, now, simulation with CO_2 as operation fluid is integrated into this tool. With CO_2 as an operation fluid, special conditions are considered for phase change between vapor and liquid phase, multiphase flow, and formation of solid CO_2 phase – where the phase change of CO_2 would result in large heat release/gain with the environment locally, and the formation of solid CO_2 phase result in blockage in the well and erosion to the tubular string.

The integrated workflow for temperature, pressure, and stress analysis is shown in Fig. 2. The coupled transient thermal and flow modeling and simulation with CO_2 injection produces the temperature of the well components (fluid flow path, tubing, in annulus spaces, casings, cements, etc.) and pressure profile of the fluid occupied area (fluid flow path, annulus spaces with fluid). The temperature and pressure profiles may be from operation of with or without CO_2 as operation fluid in order to get the change in temperature and pressure to study their impact on the stress status of the strings. Once the stress statuses under such temperature and pressure change have been obtained, then the calculation of comprehensive safety factors (based on multiple industrial standards such as API 5C3, etc.) could be performed to determine whether the design or existing well condition is strong enough to experience the CO_2 injection operation with the planned operation parameters, or it needs to redesign or to adjust operation parameters normally for reuse of old well and depleted reservoir.

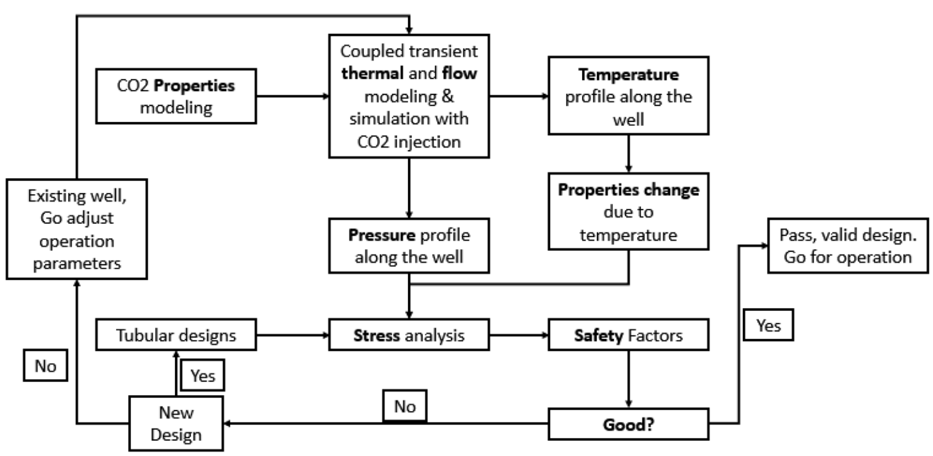

Figure 2—Integrated workflow for temperature, pressure and stress analysis

Case Study and Results Discussion

In this section, a case study is performed to study the CO_2 injection flow in well to understand their temperature and pressure behavior, and the corresponding stress behavior to evaluate the risk of the tubular strings with CO_2 injection operation. This well is an oil/gas production well to be reused for CCUS storage purpose. Fig. 3 shows the well schematic and it is a horizontal well as shown in Fig. 4.

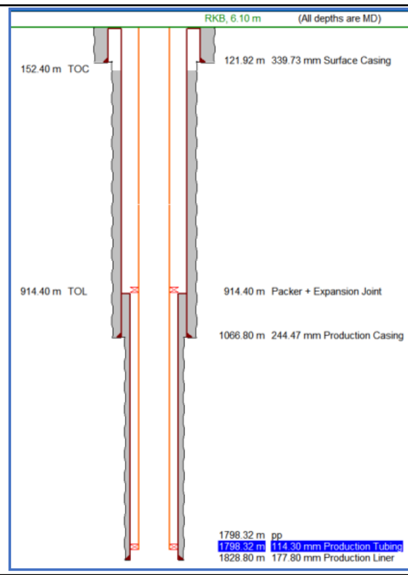

Figure 3—Integrated workflow for temperature, pressure and stress analysis

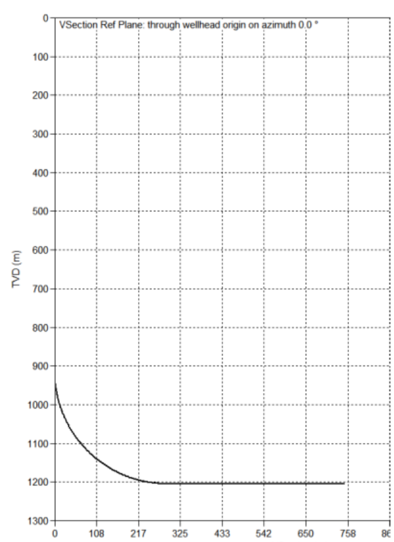

Figure 4—Integrated workflow for temperature, pressure and stress analysis

Temperature and Pressure Behavior

For improving injection efficiency purpose, the CO_2 injection state at wellhead should be in liquid phase per study of Samaroo, et al. (2024), so in the case study, the wellhead injection temperature is 2.22 °C (36 °F) and the pressure is 13789.51 kPa (2000 psi), which is in liquid phase as shown in Fig. 5. The injection rate of mass is 181.4369 Tonne/day (200 Ton/day). The perforation depth is 1828.8 m (6000 ft).

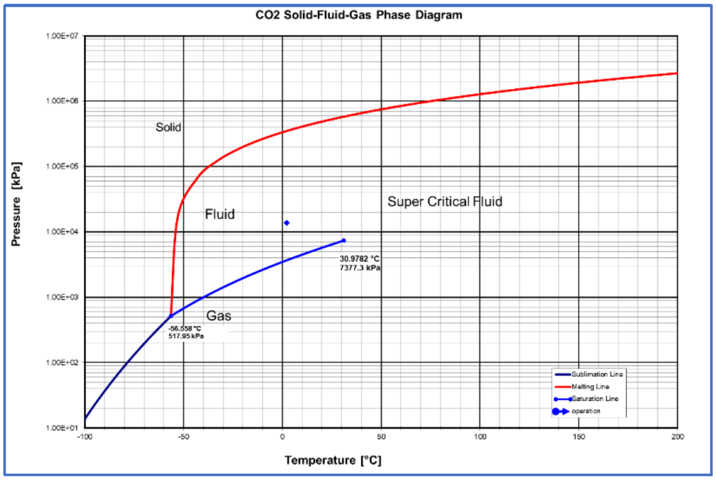

Figure 5—Wellhead injection temperature and pressure operation point

Multiple injection operations have been performed with different duration time, which is 1 day, 1 week, 1 month, 6 months and 1 year. A 1-year production operation of oil and gas is also applied for comparison purposes. Fig. 6 shows the temperature profiles of the CO_2 injection with different injection durations, the production temperature, and the geothermal. Clearly, a production operation heats the wellbore, while a CO_2 injection significantly cools the wellbore—the longer the injection, the more the well is cooled down. It is also observed that at the beginning, the well is cooled down quicker, then it is slowed down. From Fig. 6, it tells that after injection 1 day, the well temperature drops from geothermal temperature to temperature of 1 day injection is much bigger than the temperature change from injected 6 months to 1 year. The temperature history in Fig. 7 at the perforation depth gives a clear decreasing trend of the temperature behavior, which is because of pumping cooled CO_2 into the well, the longer it pumped, the lower the temperature will be. These results give quantitative value of the temperature changes.

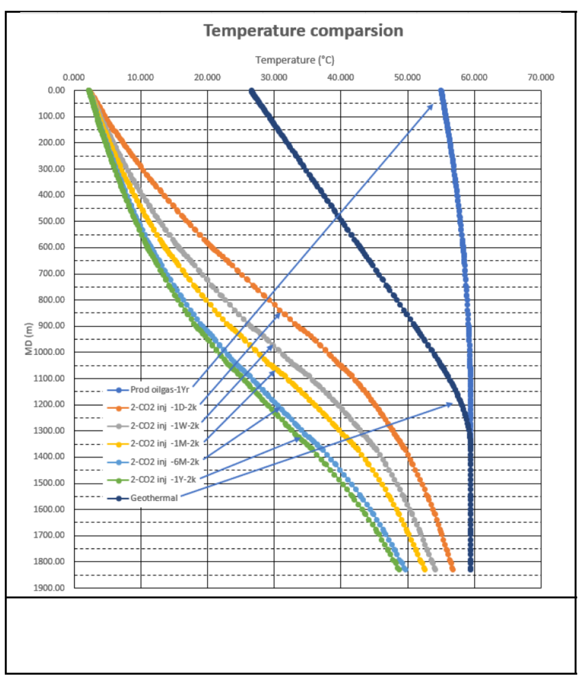

Figure 6—Temperature comparison between different operations

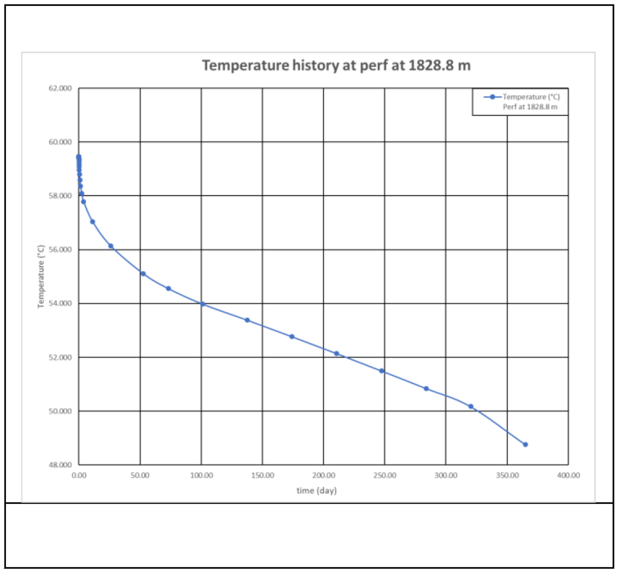

Figure 7—Temperature history at the perforation depth for 1 year CO_2 injection

Fig. 8 shows the pressure comparison between these operations. Compared with temperature difference, the difference in pressure is not that large, but there is still some trend that could be observed. The longer the injection time, the higher the bottom hole pressure is, but the increase of the bottom hole pressure is not large. Also, the reservoir is a depleted reservoir, so the bottom hole pressure of CO_2 injection could be lower than the production well bottom hole pressure to still pump the CO_2 into the formation. By coupling viewing with the temperature results, the increase of the pressure is because the decreased temperature results in the increase of the density, so the pressure is increased. The pressure history at the perforation depth for 1 year injection is shown on Fig. 9.

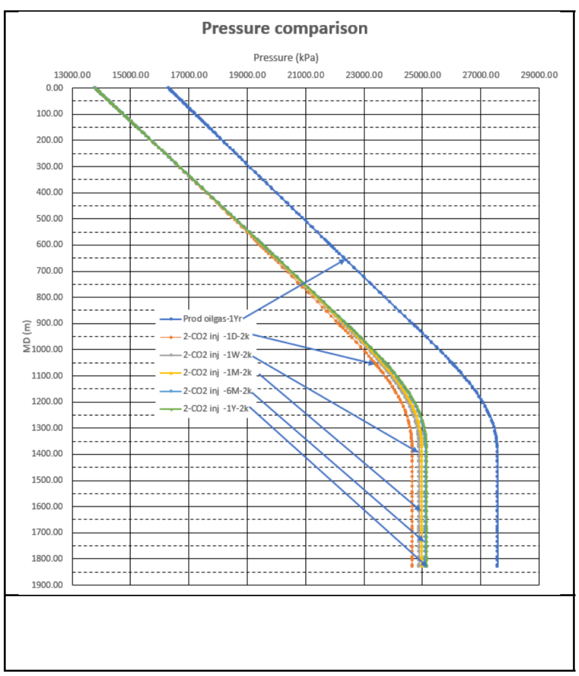

Figure 8—Pressure comparison among different operations

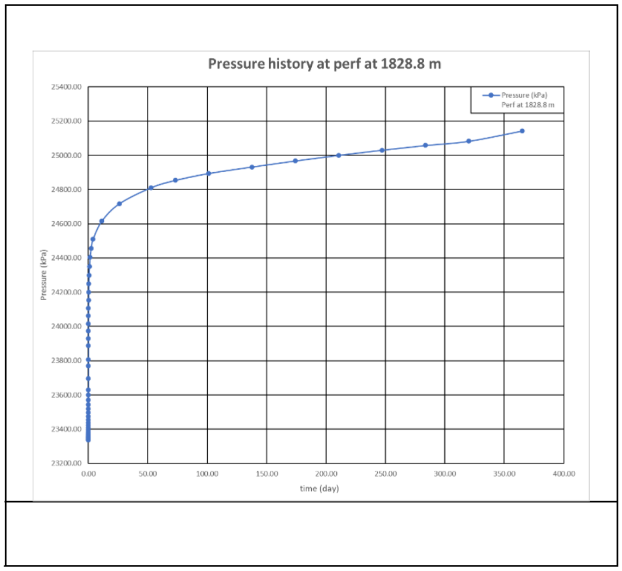

Figure 9—Pressure history at the perforation depth for 1 year CO_2 injection

The phase behavior is also important for an engineer to plan the operation. Fig. 10 shows the phase profile of the CO_2 in the well for 1 year injection. Based on the plot, the CO_2 keeps in liquid phase then enters super critical fluid phase at the bottom hole, which is what we wanted as it needs to keep the CO_2 in liquid or super critical phase to have the best efficiency of operation.

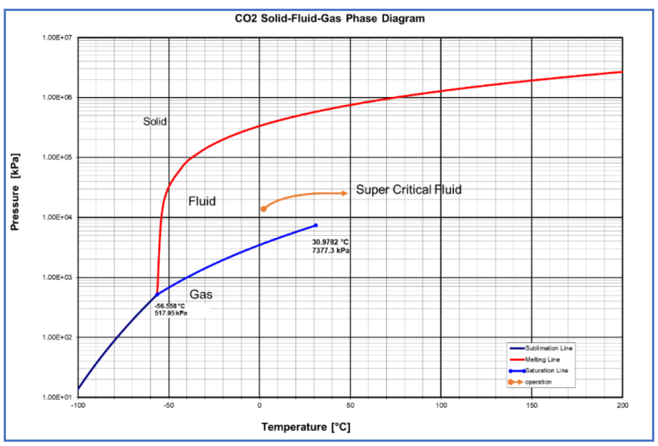

Figure 10—CO_2 phase profile along the well

Stress and Safety Analysis

Once the temperature and pressure profiles of a CO_2 injection are obtained, it is then focused on how the temperature and pressure will impact the stress applied on the strings, and whether the strings are safe. The safety factors that are based on different industrial standards are calculated to verify the safety of the strings. Fig.11 series shows the safety factors of the tubing at the initial condition when the strings are just installed, at 1 year production operation of oil and gas, and at different injection operation durations of 1 day, 1 month, and 1 year. It is observed that tubing is safe for all the safety factors, as shown in Fig. 11 (a); and it is safe after 1 year production as well, as shown in Fig. 11 (b). For CO_2 injection operation, after 1 day injection, as shown in Fig. 11 (c), the safety factors are reduced when compared to initial condition and after 1 year oil & gas production. After 1 month injection, the axial and triaxial safety factors are less than their design factors as shown in Fig. 11 (d), which means the tubing is not safe anymore. With further injection duration of 1 year, the conditions are worser, as shown in Fig. 11 (e). Why, when the CO_2 injection begins, the well is cooling down, this results in the shrinkage of the strings and induces tensile load. The behavior is further confirmed via the comprehensive design limit plot in Fig. 12 where the stress statuses change to more tensile and more burst conditions when the tubing experiences from initial condition to CO_2 injection operations. To compare the stress status between the oil & gas production and CO_2 injection, it is observed that the burst pressure difference is reduced while the tensile stress is increased which is due to cooling down induced shrinkage that results in tensile stress over the tubing.

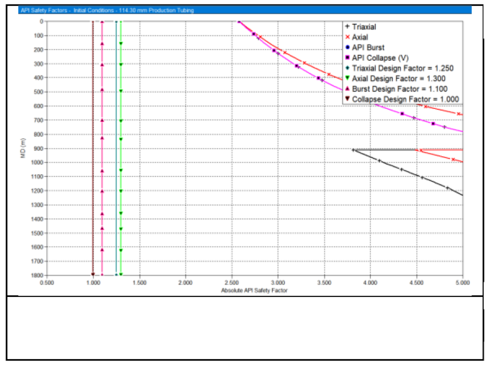

Figure 11—(a) Tubing safety factors, at initial condition

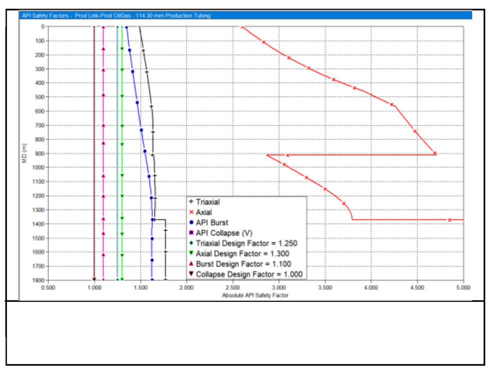

Figure 11—(b) Tubing safety factors, after 1 year oil & gas production

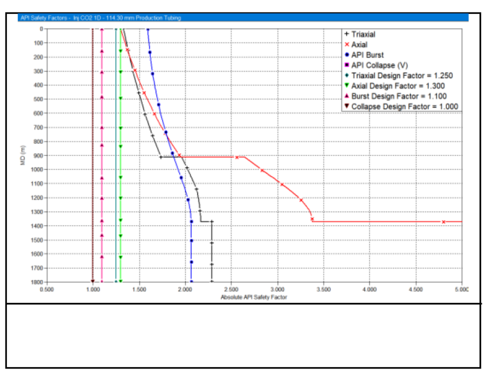

Figure 11—(c) Tubing safety factors, after CO_2 injection 1 day

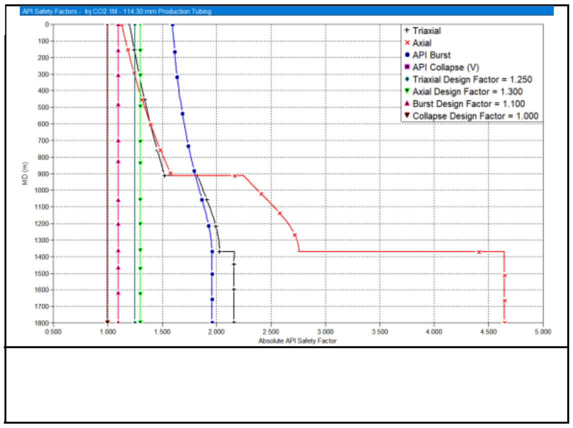

Figure 11—(d) Tubing safety factors, after CO_2 injection 1 month

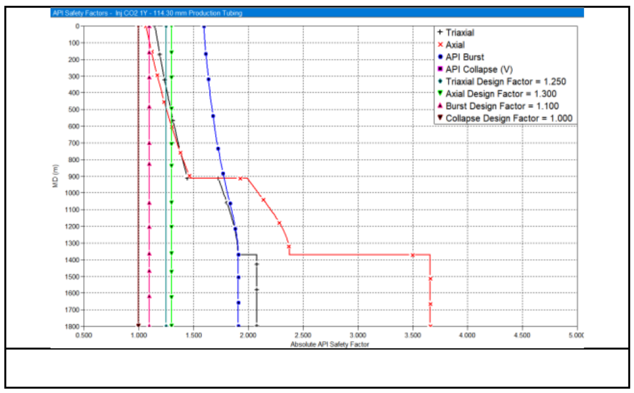

Figure 11—(e) Tubing safety factors, after CO_2 injection 1 year

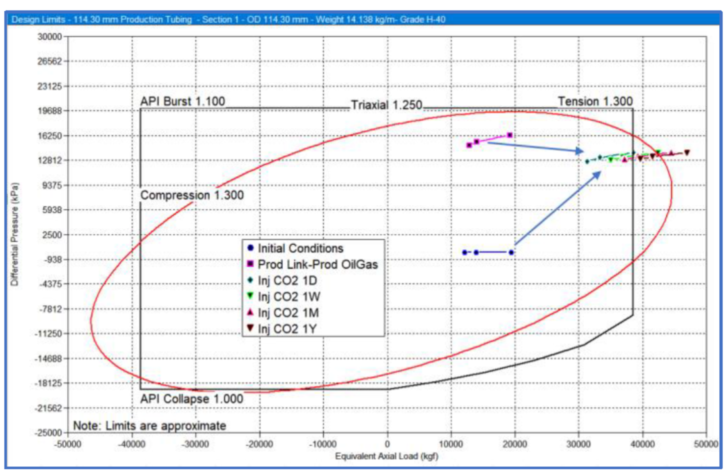

Figure 12—Comprehensive design limit plot for tubing over all the studied operations

Other Discussion

Since it is to reuse an existing well, the only thing we can do is to adjust the operation parameters. Here we tried to keep the other parameters the same but just reduce the wellhead injection pressure to be 6894.755 kPa (1000 psi) which is half of the pressure in above scenario, then the tubing is safe for all the studied CO_2 injection operations.

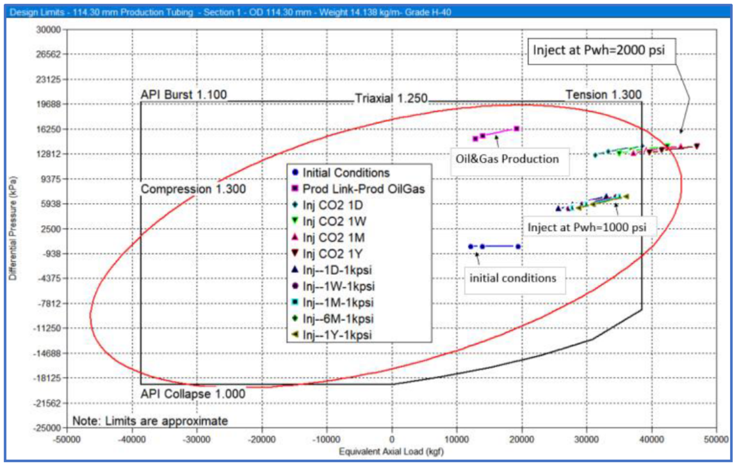

Figure 13—Comprehensive design limit plot for tubing for wellhead injection as 6894.755 kPa and the above studied operations

For this case study, it only studies the temperature and pressure behavior of CCUS CO_2 injection operation and their impact on stress status to the safety factor of the tubing. It may have some other aspects that are not considered, such as whether cooling the near well formation will result in wellbore stability issue in openhole and how, whether the cooling on the cement in annulus will lead to leaking or not and how, etc. Such phenomena would be interesting topics to study.

Conclusion

With this study, we could obtain the following conclusions:

- CO_2 injection cools down the well—CO_2 is normally injected at low temperatures and high pressure to reach in liquid phase (or super critical phase if temperature is not low enough) for best operation efficiency and handling convenience.
- The bottom hole temperature decreases, and pressure builds up with increased injection duration. This study gives a quantitative estimate of the temperature and pressure profiles and changes over time.
- Tensile failure could be expected with increased CO_2 injection duration due to cooling of the well as it would result in string material shrinkage and finally apply tensile force to the tubular strings.
- A new well design approach is needed to consider such effects for a robust well design, especially for new well that considers production and future CCUS CO_2 injection reusage.

Acknowledgement

The authors express their gratitude to Halliburton for granting permission to publish this study.

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